Talk+workshop proposal

iMAL, Center for Digital Culture and Technology

A talk and a hands-on workshop about neuro-technology and art, with a focus on the *Emotiv* software and hardware.

We will start the talk with an introduction on EEG signal and how it can be manipulated to perform different kinds of data visualizations and sonifications. We wll subsequently introduce different techniques and protocols to design real-time EEG-based interfaces, and we'll speak about our seminal audiovisual performace *Sounds of Complexity* (2005) . After that we'll introduce our most recent EEG based installation *Unstable Empathy* (2010) and talk about its conceptualization and production.

During the workshop we'll discuss about the Emotiv SDK and hardware, and we will demonstrate how to interface it with the *Max/Msp/Jitter* and *VVVV* environments. During the second part of the workshop, participants will design their own audiovisual tools to interface with the Emotiv technology (basic knowledge of max/msp/jitter and VVVV required).

Tentative timeline

Hands-on workshop: (ca. 6 hrs.)

- introduction to Emotiv software and hardware
- Emotiv SDK and OSC protocol
- interfacing with max/msp and VVVV.
- Basic sonification
- Build your own neuro-based audiovisual tools: videomixers and videosequencers

Talk (ca. 1.30 hrs.)

- introduction to EEG signal
- beta, alpha, delta, theta waves.
- sampling and digitalization of EEG signals
- real-time data protocols
- design interfaces for EEG-based media art projects
- Sounds of Complexity: audio-visualizing brainwaves in real-time
- Unstable Empathy: building intimate relationships with EEG technology